

Use-Case Flow Specification

Problem Statement #34: Digital Gift Card & Loyalty Point Exchange

Use Case ID	UC-04
Use Case Name	Redeem Gift Card
Primary Actor	Account Holder
Related Use Cases	«include» UC-05 Generate Voucher Code «extend» UC-06 Flag Suspicious Redemption

Preconditions

- The Account Holder is authenticated and logged into the platform.
- The Account Holder holds a non-zero balance of universal exchange credits.
- At least one digital gift card is available in the catalog for the desired denomination.

Postconditions

- Success: A single-use, cryptographically checksummed voucher code is generated, locked to the Account Holder, and the redeemed credit amount is deducted from their balance.
- Failure: No credits are deducted and no voucher is issued; the Account Holder's balance remains unchanged.

Main Success Scenario

1. Account Holder selects “Redeem Gift Card” and browses the gift card catalog (UC-03).
2. Account Holder selects a specific gift card and denomination.
3. System displays the number of exchange credits required for the selected card.
4. Account Holder confirms the redemption request.
5. System validates that the Account Holder's exchange credit balance is sufficient.
6. System includes Generate Voucher Code (UC-05): a single-use digital voucher code with a cryptographic checksum is generated.
7. System deducts the required exchange credits from the Account Holder's balance.
8. System locks the voucher code to the Account Holder's account so it cannot be redeemed elsewhere.
9. System displays a confirmation screen with the voucher code and sends an email/notification receipt.
10. Use case ends successfully.

Alternate Flow A — Insufficient Exchange Credits

Triggered at step 5 if the Account Holder's balance is lower than the required credits.

- 5a1. System displays an “Insufficient exchange credits” message.
- 5a2. System offers a shortcut to Convert Points to Exchange Credits (UC-02).
- 5a3. Use case ends without redemption; no credits are deducted.

Alternate Flow B — Suspicious Redemption Detected (extend: UC-06 Flag Suspicious Redemption)

Triggered after step 6 if the Fraud Monitoring System flags the redemption pattern (e.g., redemption value above threshold, or repeated high-frequency attempts from the same account).

- 6a1. System places the generated voucher on temporary hold instead of releasing it.
- 6a2. System routes the transaction to the Fraud Monitoring System for review.
- 6a3. Account Holder is notified that the redemption is pending manual review.
- 6a4. Use case ends pending resolution; credits remain reserved but are not finally deducted until the review clears.